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Immersive collaborative business process and extended reality-driven industrial metaverse technologies for economic value co-creation in 3D digital twin factories

JEL Classification: *E42; J33; O14*

Keywords: *immersive collaborative business process; extended reality-driven industrial metaverse; product lifecycle management; economic value co-creation; 3D digital twin factory; cyber-physical manufacturing enterprise*

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Abstract

Research background: Internet of Things devices and sensors, artificial intelligence-based digital asset trading and digital twin-based extended reality technologies, and autonomous robotic and enterprise resource planning systems can be leveraged in 3D semantic scene completion and metaverse-based commercial transactions across Internet of Things-based business environments. Distributed ledger and enterprise business technologies, shop-floor digital twin synthetic data, and 3D simulation and visualization systems configure integrated multi-physics workflows in hyper-realistic immersive industrial environments for artificial intelligence-based business value. Digital twin-based Internet of Robotic Things, robotic swarm and multi-modal machine learning algorithms (with regard to enterprise total factor productivity), and virtual and augmented reality simulation technologies are pivotal in spatial planning processes. Industrial product data and manufacturing value chain management support digital twin-based virtual factory modeling in collaborative immersive 3D visualization environments.

Purpose of the article: We show that interconnected business process management and metaverse economic organizational structures, immersive economic and entrepreneurial knowledge image-based modeling (for big data-driven product development processes), and remote autonomous equipment control and monitoring integrate digital twin-enabled 6G Tactile Industrial Internet of Things, deep reinforcement learning and image processing algorithms, and event-driven signal processing for collaborative economic value co-creation. Deep learning-based visual recognition and industrial extended reality technologies, 3D production management modeling, and Internet of Things industrial and mobile sensing networks are pivotal in production operation management, as deep learning-based multi-source data fusion assists autonomous industrial manufacturing processes across interactive 3D immersive business and synthetic manufacturing environments. Collaborative robotic cyber-physical production and generative Artificial Intelligence of Things-based systems (in terms of managerial business value), artificial intelligence-based perceptual and cognitive technologies, and spatial mapping and machine intelligence algorithms enhance manufacturing process visualization, as industrial big data sharing and interoperability are functional in 3D semantic scene completion for sustainable business and economic growth across big data-driven immersive virtual industrial manufacturing environments.

Methods: We inspected Tracxn (the Industrial Metaverse section) for the first 100 companies in terms of Tracxn score for X-corn status (i.e., Minicorn, Soonicorn, or none), total equity funding (USD), and company stage (i.e., Seed, Funding Raised, Unfunded, Public, Acquired, Acqui-Hired, and Series A, B, C, D), and identified three main topics for analysis that would lead to tangible business outcomes. We examined the performance management of shop floor virtualization: connected digital twins increase production and logistics process optimization in production environments across the industrial metaverse, facilitating photorealistic production system 3D modelling and simulation. We appraised integrated diagnostic functionalities of real-time simulation implementation for error elimination and machine parameter adjustment in immersive planned production lines by synthetic image data sets and collaborative workflows. We determined digital twin-based data synthesis operational procedures and interconnected use cases across industrial scalable infrastructures for value chain efficiency.

Findings & value added: We identified the specific integrated operational simulation functions and production tasks, key performance indicators of shop floor autonomous and value creation systems, and industrial process parameters for predictive quality and fault detection, resulting in production loss reduction by use of industrial metaverse technologies. By use of

operational data with regard to the technological management of the selected companies, quantitative analysis determines how immersive collaborative business process and extended reality-driven industrial metaverse technologies lead to economic value co-creation across 3D digital twin factories and cyber-physical manufacturing enterprises. The main value added derived from our research is that cloud-based collaborative 3D visualization and neuromorphic computing systems, 6G sensing and holographic simulation technologies, and machine intelligence and environment awareness algorithms (for business performance and productivity) can be leveraged in machine vision-based defect prediction, detection, diagnosis, and management. Virtual reality space convergence and object connection operate in Internet of Things-based sensing device performance monitoring across Internet of Things-based business environments. Virtual assembly lines and manufacturing enterprises necessitate machine learning-based production forecasting techniques, 3D object detection and tracking, and industrial autonomous and cyber-physical production systems, supporting spatial computing and predictive maintenance algorithms in collaborative immersive virtual environments.

Introduction

Machine vision-based defect prediction, detection, diagnosis, and management necessitate computer-vision based robotic navigation and industrial autonomous systems for enterprise resource coordination. 3D object rendering and machine intelligence algorithms and 3D immersive virtual reality and industrial metaverse technologies (in relation to performance management operations) support collaborative immersive 3D virtual environments for sustainable business growth and economic value co-creation. The economic and business management of digital twin synthetic data, multi-modal perception fusion and federated Internet of Things systems, and collaborative business process and product virtualization technologies (for production and workflow management operations) enables industrial process automation. This leads to economically viable generative artificial intelligence-based digital twin architectures of real-time production process monitoring in 3D digital twin factories. Our research is developed on the basis of literature indicating that real-time production process monitoring integrates machine intelligence and context-aware deep reinforcement learning algorithms (with regard to enterprise total factor productivity), Web3-based operation management and immersive 3D virtualization systems, and 3D computer graphics and reconfigurable intelligent surface technologies in 3D digital twin factories (Bhattacharya *et al.*, 2023; Chowdhury, 2023; Koohang *et al.*, 2023; Liu *et al.*, 2023; Starly *et al.*, 2023; Wang *et al.*, 2022a, 2022b). The research problem of this paper was whether production process simulation, digital twin-based Internet of Robotic Things, im-

age generation and visual recognition algorithms (for operation data management), and industrial metaverse production and edge computing technologies (for big data-driven product development processes) further Internet of Things-based sensing device performance monitoring for sustainable business and economic growth in cyber-physical manufacturing enterprises, shaping the chief value for business practice and theoretical managerial science.

The objective of the paper is to clarify that virtual reality object connection and space convergence necessitates extended reality and digital twin technologies (for business decision automation), industrial digital twin and product lifecycle management systems, and situational awareness and multi-sensor fusion mapping algorithms (in terms of managerial business value) in cloud-edge collaborative computing and virtual 3D environments for immersive collaborative business processes. The research gap diagnosis identifies a lack in the literature with regard to how 6G artificial intelligence-driven and industrial extended reality technologies (for business performance and productivity), metaverse economic and business systems, and quantum neural network and multi-modal machine learning algorithms can be harnessed in distributed Internet of Things-, cloud-, and digital twin-based industrial automation across interconnected immersive virtual environments, driving business and economic value co-creation.

This is the first research displaying how manufacturing process remote monitoring and diagnosis develop on production simulation data, digital twin-based virtual factory modeling, and simultaneous localization and mapping algorithms (for operation data management), particularly in relation to 6G wireless sensor and cognitive manufacturing networks in collaborative digital twin-based extended reality and virtual industrial environments. The current paper improves the state of the art by indicating that environment remote sensing augmentation, federated reinforcement and distributed machine learning algorithms (for enterprise resource coordination), and predictive maintenance simulation configure product lifecycle management due to spatial planning processes and realistic 3D representations of business transactions in the virtual economy in extended reality-driven industrial metaverse. The manuscript is organized as following: literature review (Section 2), methodology (Section 3), analysis of operations carried out by selected companies (Section 4), discussion (Section 5), and conclusions (Section 6).

Literature review

Virtual assembly lines and manufacturing enterprises develop on shop-floor digital twin synthetic data, blockchain-based federated learning and metaverse economic systems (for big data-driven product development processes), and 6G sensing and artificial intelligence-based industrial robotic technologies (Alimam *et al.*, 2023; Hou *et al.*, 2024; Netland *et al.*, 2023; Zhou *et al.*, 2023) in 3D digital twin factories for immersive collaborative business processes. Cloud-based manufacturing execution and enterprise resource planning systems, 3D factory simulation and digital twin-based semantic modeling technologies (for production and workflow management operations), and deep convolutional neural network and image processing algorithms can be deployed in big data-driven industrial manufacturing processes (Andronie *et al.*, 2021; Jagatheesaperumal & Rahouti, 2022; Özkal *et al.*, 2024) in industrial metaverse virtual spaces. With regard to business and economic relevance, object recognition and deep learning forecasting algorithms (for business performance and productivity), cyber-physical production and industrial digital twin systems, and deep learning-based visual recognition and industrial metaverse technologies (for enterprise resource coordination) enhance interconnected business process management and generative artificial intelligence-based digital twin modeling (Andronie *et al.*, 2023a; Jaimini *et al.*, 2022; Qian *et al.*, 2023) in cloud-based business environments. Decision tree and random forest algorithms (in terms of managerial business value), machine learning-based production forecasting techniques, 3D scene generation and geospatial mapping systems (in relation to performance management operations), and business process simulation and immersive geovisualization tools configure remote autonomous equipment control and monitoring (Andronie *et al.*, 2023b; Ji *et al.*, 2023; Rejeb *et al.*, 2023) for economic value co-creation in big data-driven immersive virtual industrial manufacturing environments, leading to tangible business outcomes.

The economic and business management of computer vision and support vector machine algorithms, 3D imaging sensor devices, and deep-learning-based visual Industrial Internet of Things and industrial metaverse technologies (with regard to enterprise total factor productivity) articulates real-time 3D business decision-making process representation. Virtual factory and manufacturing modeling for industrial process control and management (Aung *et al.*, 2024; Kaarlela *et al.*, 2023; Salam *et al.*, 2023)

operates in 3D multi-scene virtual and integrated multi-physics simulation environments. Internet of Things device and sensor interconnected networks, industrial machine digital twins, simultaneous localization and mapping algorithms (for business decision automation), and enterprise business and 3D visualization technologies (for operation data management) articulate metaverse economic organizational and manufacturing factory industrial structures (Balaska *et al.*, 2023; Kaigom, 2024; Semeraro *et al.*, 2023) in multi-agent-based production digital twin and collaborative virtual manufacturing environments. Business and economic applications indicate that robotized metaverse manufacturing and immersive 3D virtualization systems further predictive control-, Internet of Things-, and digital twin-based operation management. This is why industrial product virtualization, fault prevention, prediction, and diagnosis, and manufacturing simulation modelling processes (Bhattacharya *et al.*, 2023; Koohang *et al.*, 2023; Starly *et al.*, 2023) advance multi-sensory immersive environments. Big data-driven industrial modeling and simulation technologies, product lifecycle management and robotized metaverse manufacturing systems (for production and workflow management operations), and production process simulation shape virtual 3D object mapping and representation (Camacho-Muñoz *et al.*, 2023; Kshetri, 2023; Stavroulakis *et al.*, 2022) in 3D-simulated immersive and robotic simulation environments, driving business and economic value co-creation.

3D object detection and tracking, digital twin-enabled Industrial Internet of Things, and federated reinforcement and distributed machine learning algorithms (for big data-driven product development processes) further industrial process automation. Real-time 3D equipment operation and maintenance visualization sets up manufacturing factory industrial structures (Chai *et al.*, 2023; Lăzăroiu *et al.*, 2022; Stothard, 2023) in immersive collaborative virtual environments. Equipment failure prediction requires spatial mapping and deep multi-agent reinforcement learning algorithms (in terms of managerial business value), visual perception augmentation and industrial enterprise management systems (for business performance and productivity), and virtual and augmented reality simulation technologies (Chang *et al.*, 2022; Lee & Kundu, 2022; Tan *et al.*, 2023) for digital twin-enabled 6G Tactile Industrial Internet of Things in the metaverse business ecosystem. With regard to business and economic relevance, generative Artificial Intelligence of Things-based and virtual enterprise operation management systems (in relation to performance manage-

ment operations), extended reality digital twin and cloud computing technologies (for operation data management), and multi-sensor fusion mapping and artificial intelligence computer vision algorithms (with regard to enterprise total factor productivity) enhance industrial product data and manufacturing value chain management (Chen *et al.*, 2023a; Li *et al.*, 2023a; Tlili *et al.*, 2023) in collaborative immersive 3D visualization environments for metaverse business value augmentation for immersive collaborative business processes. Visual remote sensing devices, image recognition and graph convolutional network algorithms, and Internet of Things-based geospatial and reconfigurable intelligent surface technologies are pivotal for artificial intelligence-based business value (Chen *et al.*, 2023b; Li *et al.*, 2023b; Wan *et al.*, 2023) across 6G mobile wireless communication and factory automation networks for economic value co-creation in collaborative extended reality and digital twin-based industrial production environments.

Deep convolutional and wireless network-, cloud-, and computer vision-based machine intelligence depends on Internet of Things industrial and mobile sensing networks (for business decision automation), as predictive maintenance and robotic swarm algorithms, and spatial big data processing and analytics are instrumental (Chowdhury, 2023; Liu *et al.*, 2023; Wang *et al.*, 2022a, 2022b) in virtual business and enterprise operations across immersive spatial environments. Cloud-based Industrial Internet of Things operating and smart manufacturing metaverse systems, extended reality and digital twin technologies (for enterprise resource coordination), and deep learning-based multi-source data fusion assist knowledge-based enterprise organizational and autonomous industrial manufacturing processes, as production operation management and real-time production process data interoperability (Dolgui & Ivanov, 2023; Ma *et al.*, 2023; Wang *et al.*, 2024) design cloud-based extended reality environments. 3D immersive virtual reality and deep-learning-based visual Industrial Internet of Things technologies (for production and workflow management operations), robotized metaverse manufacturing and machine vision systems, and robotic swarm and object recognition algorithms enable spatial planning processes, expediting anomaly detection monitoring and real-time industrial big data visualization (Faraboschi *et al.*, 2023; Maier *et al.*, 2024; Wang *et al.*, 2022a, 2022b) in immersive 3D virtual reality environments. Environment awareness and machine vision algorithms, digital factory twins, and production machine learning and fault diagnosis systems can be

leveraged in interconnected enterprise performance and production line management (Ferrigno *et al.*, 2023; Mosco, 2023; Xiang *et al.*, 2024) in collaborative immersive virtual environments.

Method

We inspected Tracxn (the Industrial Metaverse section) for the first 100 companies (Table 1) in terms of Tracxn score for X-corn status (i.e., Minicorn, Soonicorn, or none), total equity funding (USD), and company stage (i.e., Seed, Funding Raised, Unfunded, Public, Acquired, Acqui-Hired, and Series A, B, C, D), and identified three main topics for analysis in relation to the industrial process control and management of enterprise business technologies that would lead to tangible business outcomes. By use of operational data with regard to the technological management of the selected companies, quantitative analysis determines how immersive collaborative business process and extended reality-driven industrial metaverse technologies lead to economic value co-creation across 3D digital twin factories and cyber-physical manufacturing enterprises. We examined the performance management of shop floor virtualization: connected digital twins increase production and logistics process optimization in production environments across the industrial metaverse, facilitating photorealistic production system 3D modeling and simulation in relation to i) 3D data visualization-based product planning and manufacturing processes in virtual factories, ii) the organizational structure of extended reality-based production machines and systems, iii) artificial intelligence-based forecasting capabilities for industrial collaborative value creation, planning, and processes in virtual product development based on synthetic manufacturing data across realistic immersive metaverse environments, and iv) remote computer vision-based plant maintenance, monitoring, and control. We identified the specific integrated operational simulation functions and production tasks, key performance indicators of shop floor autonomous and value creation systems, and industrial process parameters for predictive quality and fault detection, resulting in production loss reduction by use of industrial metaverse technologies. We appraised integrated diagnostic functionalities of real-time simulation implementation for error elimination and machine parameter adjustment in immersive planned production lines by synthetic image data sets and collaborative workflows. We determined

digital twin-based data synthesis operational procedures and interconnected use cases across industrial scalable infrastructures for value chain efficiency.

Topic 1.

Metaverse-based commercial transactions, virtual production planning processes, and anomaly detection monitoring require remote sensing images. Realistic production and manufacturing process virtualization and spatial big data processing and analytics (for business decision automation) make possible multi-agent-based production digital twin and collaborative virtual manufacturing environments for immersive collaborative business processes. Quantum federated learning and spatial computing algorithms (for big data-driven product development processes), smart factory network and distributed ledger technologies (with regard to enterprise total factor productivity), and real-time 3D equipment operation and maintenance visualization shape metaverse business operations and processes. Integrated multi-physics workflows support virtual manufacturing enterprise simulation and modeling in hyper-realistic immersive industrial environments. The economic and business management of Internet of Things devices and sensors, artificial intelligence-based digital asset trading and digital twin-based extended reality technologies (in relation to performance management operations), and autonomous robotic and enterprise resource planning systems (for business performance and productivity) can be leveraged in 3D semantic scene completion and metaverse-based commercial transactions across Internet of Things-based business environments. Holographic simulation and distributed ledger technologies (for production and workflow management operations), digital twin synthetic data, and metaverse economic and business systems shape real-time 3D business decision-making process representation. Industrial product data and manufacturing value chain management organize 3D production management modeling in collaborative immersive virtual environments, driving business and economic value co-creation.

Topic 2.

Virtual business and enterprise operations, manufacturing factory industrial structures, and product lifecycle management require 3D immer-

sive virtual reality and smart factory network technologies. Decision tree and random forest algorithms (for operation data management) and visual perception augmentation and geospatial mapping systems (in terms of managerial business value) give structure to 3D multi-scene virtual and integrated multi-physics simulation environments. Web3-based operation management and industrial intelligence systems (for enterprise resource coordination), 6G digital twin and vision-based sensor networks, and virtual robotic simulation and 3D computer graphics technologies can be harnessed in product life cycle operation in immersive 3D virtual reality environments. Interconnected business process management and metaverse economic organizational structures, immersive economic and entrepreneurial knowledge image-based modeling, and remote autonomous equipment control and monitoring integrate digital twin-enabled 6G Tactile Industrial Internet of Things. Deep reinforcement learning and image processing algorithms facilitate event-driven signal processing for collaborative economic value co-creation. With regard to business and economic relevance, cloud-based collaborative 3D visualization and neuromorphic computing systems, 3D digital twin visual representation and modeling, and industrial big data sharing and interoperability enable real-time 3D scene representation and reconstruction. Remote sensing images and event-driven signal processing expedite virtual reality space convergence and object connection in virtual manufacturing enterprises.

Topic 3.

Holographic interactive and mobile edge computing technologies, deep learning computer vision and artificial neural network forecasting algorithms (in terms of managerial business value), and 6G mobile communication and cognitive manufacturing networks optimize synthetic industrial image data augmentation and collaborative economic value co-creation in smart manufacturing metaverse environments so as to lead to tangible business outcomes. Realistic production and manufacturing process virtualization, virtual equipment simulation and real-time object detection systems (for big data-driven product development processes), and real-time industrial big data visualization optimize industrial equipment diagnosis and monitoring. Remote collaborative manufacturing and virtual production planning processes are critical in interactive 3D immersive business and synthetic manufacturing environments. Business and economic appli-

cations indicate that cloud-based collaborative 3D visualization and neuromorphic computing systems (for business decision automation), 6G sensing and holographic simulation technologies (in relation to performance management operations), and machine intelligence and environment awareness algorithms (for business performance and productivity) can be leveraged in machine vision-based defect prediction, detection, diagnosis, and management. Virtual reality space convergence and object connection are concatenated for Internet of Things-based sensing device performance monitoring in Internet of Things-based business environments. 6G mobile wireless communication and factory automation networks (for production and workflow management operations), extended reality digital twin and cloud computing technologies (with regard to enterprise total factor productivity), and workflow simulation and computer-vision based robotic navigation systems (for operation data management) are instrumental in predictive control-, Internet of Things-, and digital twin-based operation management in big data-driven immersive virtual industrial manufacturing environments for immersive collaborative business processes.

Results: Analysis of operations carried out by selected companies

Digital twin-enabled Industrial Internet of Things, 3D simulation and visualization systems, and situational awareness and federated reinforcement learning algorithms can be deployed in immersive economic and entrepreneurial knowledge image-based modeling. Internet of Things device failure detection paves the way for digital twin-based autonomous robotic navigation in 6G extended reality-enabled metaverse environments. Distributed ledger and enterprise business technologies, shop-floor digital twin synthetic data, and 3D simulation and visualization systems configure integrated multi-physics workflows in hyper-realistic immersive industrial environments for artificial intelligence-based business value. With regard to business and economic relevance, Internet of Things-based geo-spatial and holographic interactive technologies (for enterprise resource coordination), image generation and visual recognition algorithms, and real-time object detection and federated Internet of Things systems configure digital twin-based autonomous robotic navigation. Production process simulation lays the foundations for manufacturing factory virtualization in immersive multi-user 3D holographic environments and extended reality-driven in-

dustrial metaverse. Virtual enterprise operation management and machine vision systems (with regard to enterprise total factor productivity), spatial mapping and deep convolutional neural network algorithms, and digital twin-based semantic modeling and industrial metaverse technologies further real-time industrial big data visualization. Virtual factory and manufacturing modeling smooths the path for fault prevention, prediction, and diagnosis in industrial metaverse virtual spaces for economic value co-creation.

Internet of Things device and sensor interconnected networks, generative artificial intelligence-based industrial brains, and industrial metaverse production and edge computing technologies (in terms of managerial business value) enhance economically viable generative artificial intelligence-based digital twin architectures. Equipment failure prediction entails production process simulation for collaborative manufacturing enterprises in interconnected immersive virtual environments. Generative artificial intelligence-based industrial brains, situational awareness and deep reinforcement learning algorithms, and 3D computer graphics and holographic communication technologies (for production and workflow management operations) assist production process visualization for collaborative manufacturing enterprises in machine intelligence economics with regard to 3D object detection and tracking. The economic and business management of deep learning-based visual recognition and industrial extended reality technologies (in relation to performance management operations), 3D production management modeling, and Internet of Things industrial and mobile sensing networks (for business performance and productivity) is pivotal in production operation management. Deep learning-based multisource data fusion runs interference for autonomous industrial manufacturing processes across interactive 3D immersive business and synthetic manufacturing environments. Deep convolutional and wireless network-, cloud-, and computer vision-based machine intelligence, 3D object rendering and quantum neural network algorithms (for big data-driven product development processes), and holographic communication and virtual robotic simulation technologies (for business decision automation) enable virtual 3D object mapping and representation. Synthetic industrial image data augmentation reinforces knowledge-based enterprise organizational processes for cyber-physical manufacturing enterprises in collaborative immersive 3D virtual environments, driving business and economic value co-creation.

Remote robotic and machine operations integrate collaborative robotic cyber-physical production and workflow simulation systems (for operation data management), context-aware deep reinforcement learning and computer vision-based navigation algorithms, and artificial intelligence-based perceptual and cognitive technologies in virtual manufacturing factories. 6G mobile communication and vision-based sensor networks, collaborative robotic cyber-physical production and blockchain-based federated learning systems (with regard to enterprise total factor productivity), and mobile edge computing and industrial metaverse technologies can be harnessed in realistic 3D representations of business transactions in the virtual economy across robotic simulation environments. Business and economic applications indicate that virtual assembly lines and manufacturing enterprises necessitate machine learning-based production forecasting techniques, industrial autonomous and cyber-physical production systems (for enterprise resource coordination), and spatial computing and predictive maintenance algorithms for 3D object detection and tracking in collaborative immersive virtual environments. Distributed Internet of Things-, cloud-, and digital twin-based industrial automation develops on situational awareness and federated reinforcement learning algorithms, digital twin-based extended reality and 3D computer graphics technologies, and enterprise resource planning and autonomous robotic systems in 6G extended reality-enabled metaverse environments for economic value co-creation.

Collaborative business process and product virtualization technologies (for production and workflow management operations), robotized metaverse manufacturing and visual perception augmentation systems, and machine vision and image recognition algorithms (in terms of managerial business value) articulate remote collaborative manufacturing. Metaverse business operations and processes draw on virtual manufacturing enterprise simulation and modeling in 3D virtual factory environments for immersive collaborative business processes. Digital twin-based Internet of Robotic Things, robotic swarm and multi-modal machine learning algorithms (in relation to performance management operations), and virtual and augmented reality simulation technologies are pivotal in spatial planning processes. Industrial product data and manufacturing value chain management capitalizes on digital twin-based virtual factory modeling in collaborative immersive 3D visualization environments so as to lead to tangible business outcomes. Visual remote sensing devices, immersive

geovisualization tools, and big data-driven industrial modeling and simulation technologies (for big data-driven product development processes) assist environment remote sensing augmentation. Virtual reality object connection and space convergence are grounds for industrial product virtualization in virtual manufacturing factories across immersive multi-user 3D holographic environments. 3D imaging sensor devices, immersive geovisualization tools, and deep learning forecasting and graph convolutional network algorithms (for business decision automation) assist real-time 3D scene representation and reconstruction for cloud-based Industrial Internet of Things operating and smart manufacturing metaverse systems (for business performance and productivity) in hyper-realistic immersive industrial environments.

3D simulation and visualization systems, collaborative business process and product virtualization technologies (for operation data management), and multi-sensor fusion mapping and quantum neural network algorithms are instrumental in knowledge-based enterprise organizational processes. Remote collaborative manufacturing has as a precondition real-time 3D scene representation and reconstruction in industrial metaverse virtual spaces for sustainable business growth. The economic and business management of collaborative robotic cyber-physical production and generative Artificial Intelligence of Things-based systems, artificial intelligence-based perceptual and cognitive technologies, and spatial mapping and machine intelligence algorithms (with regard to enterprise total factor productivity) enhances manufacturing process visualization. Industrial big data sharing and interoperability smooth 3D semantic scene completion for sustainable business and economic growth in big data-driven immersive virtual industrial manufacturing environments for immersive collaborative business processes. Artificial intelligence-based industrial robotic and 3D factory simulation technologies, Internet of Things devices and sensors, and industrial machine digital twins optimize industrial equipment diagnosis and monitoring. Real-time production process data interoperability expedites 3D digital twin visual representation and modeling in multi-sensory immersive and 3D virtual factory environments. Business and economic applications indicate that mobile edge computing and industrial metaverse technologies, image generation and visual recognition algorithms (in terms of managerial business value), and 6G wireless sensor and 6G mobile communication networks can be harnessed in metaverse business value augmentation. Industrial process control and management establish on

manufacturing process visualization in 3D digital twin factories across collaborative digital twin-based extended reality environments, driving business and economic value co-creation.

Geospatial mapping and real-time object detection systems (for production and workflow management operations) are pivotal in industrial big data sharing and interoperability in 6G extended reality-enabled metaverse and collaborative immersive 3D virtual environments. With regard to business and economic relevance, manufacturing simulation data, generative Artificial Intelligence of Things-based and blockchain-based federated learning systems (in relation to performance management operations), and extended reality and digital twin technologies (with regard to enterprise total factor productivity) are pivotal in interconnected enterprise performance and production line management. Real-time 3D business decision-making process representation encompasses virtual factory and manufacturing modeling for collaborative manufacturing enterprises. Product lifecycle management, event-driven signal processing, and anomaly detection monitoring necessitate cloud-based collaborative 3D visualization and neuromorphic computing systems. Image processing and deep convolutional neural network algorithms (for enterprise resource coordination) and digital twin-based extended reality and industrial metaverse technologies (for business decision automation) are critical in collaborative digital twin-based extended reality and immersive collaborative virtual environments so as to lead to tangible business outcomes.

Internet of Things industrial and mobile sensing networks, deep reinforcement learning and machine intelligence algorithms (for big data-driven product development processes), and multi-modal perception fusion and metaverse economic systems (for business performance and productivity) enhance real-time production process monitoring and 3D semantic scene completion in smart manufacturing metaverse environments. Digital twin-enabled Industrial Internet of Things, decision tree and random forest algorithms (in terms of managerial business value), and 6G mobile wireless communication and cognitive manufacturing networks can be leveraged in predictive maintenance simulation for immersive collaborative business processes. Realistic production and manufacturing process virtualization fosters collaborative economic value co-creation in immersive spatial environments. Immersive economic and entrepreneurial knowledge image-based modeling integrates simultaneous localization and mapping algorithms. Industrial machine digital twins establish on artificial

intelligence-based industrial robotic and digital twin-based semantic modeling technologies (for operation data management) in 3D digital twin factories across virtual 3D environments. 3D scene generation and industrial intelligence systems are instrumental in predictive control-, Internet of Things-, and digital twin-based operation management in collaborative extended reality and digital twin-based industrial production environments for economic value co-creation.

Discussion

We have developed our research on Ganchev *et al.* (2023), Mourtzis and Angelopoulos (2023) and Xuhong and Xuan, 2023), and clarified how robotized metaverse manufacturing and product lifecycle management systems, extended reality digital twin and cloud computing technologies, and graph convolutional network and situational awareness algorithms can be deployed in economically viable generative artificial intelligence-based digital twin architectures in 3D digital twin factories across multi-sensory immersive and virtual industrial environments for immersive collaborative business processes. Computer vision and support vector machine algorithms, digital twin-enabled 6G Tactile Industrial Internet of Things, and holographic communication and artificial intelligence-based digital asset trading technologies (with regard to enterprise total factor productivity) optimize big data-driven industrial manufacturing processes. Product life cycle operations are grounds for industrial equipment diagnosis and monitoring in the metaverse business ecosystem across smart manufacturing metaverse environments. Related previous research focused mainly on how cloud-based manufacturing execution and enterprise resource planning systems (for operation data management), digital factory twins, and machine vision and spatial computing algorithms are instrumental in remote robotic and machine operations. Manufacturing factory virtualization complements predictive maintenance simulation in 3D-simulated immersive environments. Business and economic applications indicate that cyber-physical production and workflow simulation systems configure generative artificial intelligence-based digital twin modeling and interconnected business process management in virtual manufacturing enterprises across collaborative immersive 3D visualization and cloud-based extended reality environments.

Our findings complement recent research (e.g., Gourisetti *et al.*, 2023; Mourtzis *et al.*, 2023; Yi *et al.*, 2023) with regard to how deep multi-agent reinforcement learning and computer vision-based navigation algorithms (for business decision automation), artificial intelligence-based perceptual and cognitive technologies, and metaverse economic and industrial enterprise management systems (in relation to performance management operations) can be deployed in Internet of Things device failure detection. Manufacturing process visualization has as a necessary condition generative artificial intelligence-based digital twin modeling in the metaverse business ecosystem. Artificial neural network forecasting and quantum federated learning algorithms (for big data-driven product development processes), production machine learning and virtual equipment simulation systems (for business performance and productivity), and 6G artificial intelligence-driven and 3D visualization technologies (for production and workflow management operations) articulate virtual business and enterprise operations in cloud-based business environments. Web3-based operation management and immersive 3D virtualization systems, business process simulation tools (for enterprise resource coordination), and virtual and augmented reality simulation technologies (in terms of managerial business value) articulate real-time production process data interoperability. Manufacturing factory virtualization is decisive in digital twin-based autonomous robotic navigation across virtual industrial environments and for metaverse-based commercial transactions. The economic and business management of machine intelligence and object recognition algorithms, fault diagnosis and robotized metaverse manufacturing systems, and 3D factory simulation and holographic interactive technologies can be leveraged in virtual reality space convergence and object connection. Metaverse business operations and processes and realistic 3D representations of business transactions are crucial in the virtual economy for virtual manufacturing enterprises across 3D multi-scene virtual and integrated multi-physics simulation environments for economic value co-creation.

We identified similar results with Guo *et al.* (2023), Nagy *et al*. (2023) and Yu *et al.* (2023) in terms of how artificial intelligence computer vision and multi-modal machine learning algorithms, 3D scene generation and industrial intelligence systems (with regard to enterprise total factor productivity), and industrial metaverse and deep learning-based visual recognition technologies shape interconnected enterprise performance and production line management. Environment remote sensing augmentation

upgrades industrial process automation in collaborative extended reality and digital twin-based industrial production environments for immersive collaborative business processes. Manufacturing process remote monitoring and diagnosis, production operation management, and Internet of Things device failure detection integrate Internet of Things devices and sensors. Smart factory network and industrial extended reality technologies (for operation data management) straighten out 6G wireless sensor and factory automation networks in immersive spatial environments. With regard to business and economic relevance, artificial intelligence-based business value in Internet of Things-based business and 3D-simulated immersive environments require computer vision and support vector machine algorithms (for business performance and productivity), autonomous robotic and production machine learning systems (in relation to performance management operations), and digital twin synthetic data. Metaverse economic organizational structures, manufacturing simulation modeling processes, and Internet of Things-based sensing device performance monitoring develop on Internet of Things device and sensor interconnected networks. Federated Internet of Things and fault diagnosis systems and deep multi-agent reinforcement learning and environment awareness algorithms shape up cyber-physical production enterprises in multi-agent-based production digital twin and collaborative virtual manufacturing environments, driving business and economic value co-creation.

Based on Han *et al.* (2023), Navarro and Pita (2023), Zeng *et al.* (2023), we showed how deep learning-based visual Industrial Internet of Things technologies (for business decision automation), industrial enterprise management and virtual equipment simulation systems, and 3D object rendering and predictive maintenance algorithms optimize manufacturing simulation modeling processes. Generative artificial intelligence-based industrial brains give structure to production process simulation in extended reality-driven industrial metaverse and immersive multi-user 3D holographic environments. Our findings support previous research (e.g., Chai *et al.*, 2023; Lăzăroiu *et al.*, 2022; Stothard, 2023) showing how fault prevention, prediction, and diagnosis, product life cycle operations, and manufacturing process visualization require cloud-based Industrial Internet of Things operating and smart manufacturing metaverse systems. Enterprise business and 6G artificial intelligence-driven technologies (for production and workflow management operations) and computer vision-based navigation and artificial neural network forecasting algorithms upgrade cloud-based

extended reality environments for sustainable business and economic growth. Business and economic applications indicate that 6G digital twin networks, deep learning forecasting algorithms, and industrial autonomous systems articulate remote sensing images. Digital twin-based virtual factory modeling supports manufacturing process visualization in immersive 3D virtual reality environments. Deep learning computer vision and context-aware deep reinforcement learning algorithms (in terms of managerial business value), Internet of Things-based geospatial and 3D computer graphics technologies (for big data-driven product development processes), and enterprise resource planning and machine vision systems assist virtual reality object connection and space convergence for immersive collaborative business processes. Autonomous industrial manufacturing processes necessitate remote autonomous equipment control and monitoring in interactive 3D immersive business and synthetic manufacturing environments so as to lead to tangible business outcomes.

Conclusions

Spatial big data processing and analytics, visual remote sensing devices, and reconfigurable intelligent surface and artificial intelligence-based digital asset trading technologies (for enterprise resource coordination) enable virtual production planning processes. 3D production management modeling straightens out integrated multi-physics workflows in cloud-edge collaborative computing environments. Barriers in, and negative aspects of, implementation comprise how federated reinforcement and distributed machine learning algorithms, industrial metaverse production and edge computing technologies (with regard to enterprise total factor productivity), and computer-vision based robotic navigation and multi-modal perception fusion systems (for operation data management) can be deployed in metaverse business value augmentation. Manufacturing process remote monitoring and diagnosis upgrade machine vision-based defect prediction, detection, and management for sustainable business growth in machine intelligence economics. The economic and business management of 6G digital twin and vision-based sensor networks (for business performance and productivity), big data-driven industrial modeling and simulation technologies (in relation to performance management operations), and 3D imaging sensor devices furthers industrial product virtualization. Virtual

manufacturing enterprise simulation and modeling pave the way for equipment failure prediction in virtual assembly lines and manufacturing enterprises across cloud-edge collaborative computing and virtual 3D environments. Business and economic applications indicate that deep convolutional and wireless network-, cloud-, and computer vision-based machine intelligence (for big data-driven product development processes) develops on digital twin-based Internet of Robotic Things. Business process simulation tools and machine learning-based production forecasting techniques drive forward virtual manufacturing factories across cloud-based business environments for economic value co-creation.

Organizational and managerial challenges encompass how industrial metaverse and 6G sensing technologies, manufacturing simulation data, and situational awareness and federated reinforcement learning algorithms (for business decision automation) shape real-time 3D equipment operation and maintenance visualization. Industrial process control and management necessitate 3D digital twin visual representation and modeling in machine intelligence economics across interconnected immersive virtual environments so as to lead to tangible business outcomes. Future research should further investigate whether synthetic industrial image data augmentation, virtual 3D object mapping and representation, and big data-driven industrial manufacturing processes necessitate artificial intelligence computer vision and quantum federated learning algorithms. 3D visualization and holographic simulation technologies (for production and workflow management operations) support cloud-based manufacturing execution and enterprise resource planning systems in 3D virtual factory environments. Limitations include a certain platform (Tracxn) for startup and private company tracking, and our analysis covers only the first 100 companies integrating virtual robotic simulation and 3D computer graphics technologies (in terms of managerial business value), metaverse economic and business systems, and digital factory twins. Production process simulation builds up remote robotic and machine operations and deep learning-based multi-source data fusion in immersive collaborative virtual environments. Policy implications include how deep learning computer vision and image recognition algorithms (in relation to performance management operations), industrial digital twin and virtual enterprise operation management systems, and shop-floor digital twin synthetic data can be harnessed in distributed Internet of Things-, cloud-, and digital twin-based industrial

automation in robotic simulation environments, driving business and economic value co-creation.

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Data availability statement

All data generated or analyzed are included in the published article. The raw data supporting the conclusion of this article will be made available by the authors,

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Author contributions

All listed authors have made a substantial, direct and intellectual contribution to the work, and approved it for publication. The authors take full responsibility for the accuracy and the integrity of the source analysis.

Conflict of interest statement

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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Annex

Table 1. Companies under research

No.	Company name	Tracxn score	Description	X-corn status	Total equity funding (USD)	Company stage	Investor name
1.	Matterport	83	Provider of cloud and AI-based virtual tour solutions for construction professionals	-	163M	Acquired	Institutional: Qualcomm Ventures
2.	Attensi	69	Provider of gamified simulation training for corporates	Minicorn	32.3M	Series B	Institutional: Viking Venture
3.	OpenSpace	67	SaaS-driven site monitoring solution for construction projects	Soonicorn	199M	Series D	Institutional: Taronga Ventures
4.	Bentley Systems	66	Provider of cloud-based designing software for the construction industry	-	-	Public	Institutional: Argosy Capital
5.	Strivr	64	Virtual reality-based LMS platforms	Soonicorn	86M	Series B	Institutional: Workday Ventures
6.	Atheer	62	Provider of an augmented reality-based platform for enterprise applications	Minicorn	36.8M	Series C	Institutional: RONA Holdings
7.	XYZ Reality	62	Provider of cloud-based augmented reality software for construction professionals	Minicorn	34.5M	Series A	Institutional: Octopus Ventures
8.	Cognite	61	Provider of AI-based production data analysis software	-	225M	Acquired	Institutional: TCV
9.	Librestream	60	On site video collaboration platform	Minicorn	55M	Series D	Institutional: Emerald Technology Ventures
10.	Interplay Learning	60	Provider of VR simulation-based online training solutions	Minicorn	46.1M	Series C	Institutional: Goldman Sachs

Table 1. Continued

No.	Company name	Tracxn score	Description	X-corn status	Total equity funding (USD)	Company stage	Investor name
11.	Scope AR	60	Provider of an AR-based software offering remote assistance support solution	Minicorn	11.8M	Series A	Institutional: SignalFire
12.	Help Lightning	60	Provider of an augmented reality-based remote collaboration solution	Minicorn	12.7M	Series A	Institutional: Engage
13.	Terabase	60	Developer of a web-based solar power plant optimization platform	Minicorn	77M	Series B	Institutional: Fifth Wall
14.	Augmentir	59	Provider of a platform offering AR solutions for industrial front-line workers	Minicorn	10.5M	Series A	Institutional: Lerer Hippeau
15.	Coupa	58	Provider of AI-based simulation and modelling software for supply chains	-	56.1M	Acquired	Institutional: TPG
16.	Sightcall	58	Provider of a cloud-based visual collaboration platform	Soonicorn	67M	Series B	Institutional: InfraVia Capital Partners
17.	Aker Solutions	58	Provider of diversified solutions for the oil and gas industry	-	-	Acquired	Corporate: European Union
18.	Talespin	57	Developer of an immersive platform for recruitment and training applications	-	40.2M	Acquired	Institutional: WestRiver Group
19.	Neara	57	Developer of a power transmission digital twin solution	Minicorn	62.2M	Series C	Institutional: EQT
20.	TechSee	57	AI virtual agent for customer experience	Soonicorn	53.5M	Series C	Institutional: TELUS Ventures
21.	Transfr	57	Online VR-based LMS for enterprises	Soonicorn	90M	Series C	Institutional: ABS Capital

Table 1. Continued

No.	Company name	Tracxn score	Description	X-corn status	Total equity funding (USD)	Company stage	Investor name
22.	PassiveLogic	57	Provider of digital-twin technology-based solutions for construction projects	Minicorn	80M	Series B	Institutional: nVentures
23.	Upskill	57	Provider of an enterprise integrated app platform for wearables and AR	-	53.4M	Acquired	Institutional: Cisco Investments
24.	Ubimax	56	Augmented reality based wearable picking solution	-	6M	Acquired	Institutional: Fielmann Ventures
25.	LocusView	56	Navigation-based workforce management solutions for the construction sector	Soonicorn	80M	Series A	Institutional: Israel Growth Partners
26.	SenseHawk	56	Provider of drone-based aerial survey systems and cloud-based data analysis tools	-	7.45M	Acquired	Institutional: ADQ
27.	StructionSite	55	Provider of SaaS-driven construction site photo documentation platform	-	25M	Acquired	Institutional: CEMEX Ventures
28.	Almer Technologies	55	Provide of AR-based industrial remote assistance platforms	-	11.9M	Acquired	Institutional: VI Partners
29.	Intangles	54	Digital twin and IoT-enabled solutions for vehicle monitoring and fleet management	Minicorn	11.7M	Series A	Institutional: Baring Private Equity Partners
30.	Converge	54	Sensors networks for monitoring construction fields	Minicorn	18.8M	Series A	Institutional: MassChallenge
31.	Avvir	54	Provider of inspection management software for construction sites	-	12.6M	Acquired	Institutional: Trust Ventures

Table 1. Continued

No.	Company name	Tracxn score	Description	X-corn status	Total equity funding (USD)	Company stage	Investor name
32.	Hololight	54	Diverse XR applications	Minicorn	27M	Series B	Institutional: EnBW New Ventures
33.	HoloBuilder	54	Webeditor to create VR experiences	-	2.25M	Acquired	Institutional: Foundamental
34.	DAQRI	54	AR-based workforce productivity management software	Soonicorn	15M	Series A	Institutional: Tarsadia
35.	CareAR	54	Provider of augmented reality based visual support solution	-	10M	Acquired	Corporate: ServiceNow
36.	Ario	53	Provider of AR-based productivity platform for enterprises	-	2.75M	Acquired	Institutional: NIH SBIR/STTR
37.	Tignis	53	Provider of AI-powered process control solutions	-	16.1M	Acquired	Institutional: DN Capital
38.	SiteAware	53	Cloud-based project monitoring and site management solution	Minicorn	30M	Series B	Institutional: Vertex Ventures Israel
39.	Twinner	53	Provider of a cloud-driven merchandising solution for the automotive industry	Minicorn	47.6M	Series B	Institutional: Bmp Ventures
40.	DocuSketch	53	Documentation and visualization solution for the construction industry	-	-	Series A	Institutional: Summit Partners
41.	Mursion	52	Provider of virtual reality simulations for workplace training	Soonicorn	40.6M	Series B	Institutional: Leeds Illuminate
42.	Trezi	52	Provider of SaaS-based VR-based visualization solutions	Minicorn	3.3M	Series A	Institutional: Windrose Capital
43.	Immense	52	Provides a platform for transportation simulation and intelligence	-	4.6M	Series A	Institutional: Amadeus Capital

Table 1. Continued

No.	Company name	Tracxn score	Description	X-corn status	Total equity funding (USD)	Company stage	Investor name
44.	Decisyon	52	Provider of platform for corporate performance management	Soonicorn	55.9M	Series B	Institutional: Catalyst Investors
45.	Indus AI	51	Digital twin platform for construction	-	11.7M	Acqui-Hired	Institutional: Millennium Technology Value Partners
46.	VIRNECT	51	Provider of augmented reality solutions for industries	-	37.6M	Public	Institutional: STIC Investments
47.	Doxel	51	Provider of SaaS and AI-driven construction productivity management solutions	Soonicorn	56.5M	Series B	Institutional: Insight Partners
48.	Treble	50	Provider of a VR and cloud-based sound simulation platform	Minicorn	20M	Series A	Institutional: Kompas
49.	Prodea	50	Provider of cloud-based solutions for IoT applications	Soonicorn	156M	Series D	Institutional: DCM Ventures
50.	Verses	50	Provider of AI-integrated digital twins for data distribution	-	-	Public	Corporate: G42
51.	VEERUM	50	Provider of digital twin based asset visualization tool for manufacturing units	Minicorn	23.1M	Series B	Institutional: Evok Innovations
52.	Simulanis	50	Provider of AR and VR-based training systems for multiple industries	Minicorn	1.44M	Seed	Institutional: VilCap Investments
53.	Reconstruct	50	SaaS-driven visual analytics platform for the AEC industry	Minicorn	27.4M	Series B	Institutional: Serra Ventures
54.	XMReality	49	AR-based field application for visual guidance	-	1.35M	Acquired	Institutional: Almi Invest

Table 1. Continued

No.	Company name	Tracxn score	Description	X-corn status	Total equity funding (USD)	Company stage	Investor name
55.	Streem	49	Provider of AR and computer vision-based customer engagement and support software	-	2.85M	Acquired	Institutional: GVR Fund
56.	River Logic	49	Advanced analytics platforms for prescriptive analytics	-	16.7M	Series C	Institutional: Mercury Ventures
57.	3i	49	Provider of hardwares and software solutions for building digital twins	Minicorn	24M	Series A	Institutional: LB Investment
58.	Allvision	49	Provider of an AI-based spatial data platform	-	3.2M	Seed	Institutional: Lavrock Ventures
59.	FutureOn	48	Provider of a cloud-based digital twin software solution for oilfield asset management	-	-	Acquired	Institutional: Bentley Acceleration Fund
60.	Visual Vocal	48	VR-based collaboration platform for architecture, construction and manufacturing industry	Minicorn	4.1M	Seed	Institutional: Eniac Ventures
61.	The Moon Hub	48	Cloud-based VR training platform for companies	-	4.5M	Seed	Institutional: Unconventional Ventures
62.	Diota	48	Industrial augmented reality solution for field service application	-	9.06M	Acquired	Institutional: Supernova Invest
63.	PIXO VR	47	Provider of a virtual reality-based workforce training solution	Minicorn	6.9M	Series A	Institutional: Aramco Ventures
64.	4M Analytics	47	Provider of AI-based security intelligence solutions	-	45M	Series A	Institutional: Insight Partners
65.	RE'FLEKT	47	AR platform for industrial applications & real-estate	-	10.1M	Acqui-Hired	Institutional: German Accelerator

Table 1. Continued

No.	Company name	Tracxn score	Description	X-corn status	Total equity funding (USD)	Company stage	Investor name
66.	Enscape	47	SaaS-based provider of 3D visualization solutions	-	-	Series A	Institutional: HTGF
67.	Pristine	46	Field service management Solutions through Google Glass.	-	7.43M	Acqui-Hired	Institutional: S3 Ventures
68.	Taqtile	46	Provider of an augmented and virtual reality platform for smart glasses	Minicorn	8.2M	Seed	Institutional: 5G Open Innovation Lab
69.	Light Guide Systems	46	Provider of an augmented reality tool for assembly and manufacturing processes	Minicorn	15M	Series B	Institutional: Michigan Capital Advisors
70.	Samp	46	Digital twin for the visualization of manufacturing facilities	Minicorn	10.5M	Series A	Institutional: Promus Ventures
71.	TWI	46	Provider of digital twin solution for wind farms	-	-	Funding Raised	Corporate: European Union
72.	Car360	46	Augmented reality based vehicle imagery solutions	-	3.55M	Acquired	Institutional: Service Provider Capital
73.	Fieldbit	45	Maintenance visualization system	-	2.21M	Acquired	Institutional: I2BF Global Ventures
74.	FJ Dynamics	45	Provider of robots and automated systems for multiple industries	Soonicorn	99.4M	Series B	Institutional: Greater Bay Area Homeland Investments
75.	CUSMAT	45	Cloud-based LMS platform for multi-industry	Minicorn	4.22M	Series A	Institutional: Capria
76.	Scanifly	45	Provider of a drone platform for solar system installations	Minicorn	2.27M	Seed	Institutional: Lightbank
77.	Uptale	45	VR-based learning platform for businesses	-	-	Funding Raised	Institutional: Inveready
78.	Squint	44	Provider of workflow SOPs to instruction converter	Minicorn	19M	Series A	Institutional: Sequoia Capital

Table 1. Continued

No.	Company name	Tracxn score	Description	X-corn status	Total equity funding (USD)	Company stage	Investor name
79.	Pointivo	44	AI-driven inspection maintenance service provider	Minicorn	8.38M	Series B	Institutional: VentureSouth
80.	Kavida.ai	43	Developer of AI-powered digital twin and purchase order platforms for supply chain management	-	1.57M	Seed	Institutional: redbusventures.com
81.	ALGO8	43	Provider of AI application development solutions	-	-	Seed	Institutional: Unigrowth
82.	GLARTEK	43	IoT platform for visualizing data in real time using AR-enabled apps and devices	-	1.7M	Seed	Institutional: EDP Ventures
83.	KIT-AR	43	Application providing augmented reality solutions for manufacturing industry	-	5.68M	Series A	Institutional: Criteria Venture Tech
84.	Kognitiv Spark	43	Cloud-based industrial holographic platform for remote support & training	-	3.26M	Series A	Institutional: New Brunswick Innovation Foundation
85.	Cityzenith	43	Developer of a digital twin platform for construction industries	Minicorn	11M	Series A	Institutional: Sequoia Capital
86.	Corva	42	Provider of predictive drilling analytics	-	-	Unfunded	-
87.	Skillveri	42	Online VR-based LMS platform for enterprises	Minicorn	1.21M	Series A	Institutional: Ankur Capital
88.	Sensat	42	Provider of a digital twin platform for the construction industry	Minicorn	34.8M	Series B	Institutional: National Grid Partners
89.	DesignX	42	Provider of manufacturing operations management tool	-	2.32M	Seed	Institutional: Piper Serica

Table 1. Continued

No.	Company name	Tracxn score	Description	X-corn status	Total equity funding (USD)	Company stage	Investor name
90.	TI Mining	42	IoT and digital twin based solution for mining industry	Minicorn	13.8M	Series B	Institutional: Kayyak Ventures
91.	Synergy	42	Cloud-based XR platform for collaboration, communication and remote support	-	7.73M	Series A	Institutional: Kvanted
92.	Eugenie	42	Provider of AI-based machine and process management software solution for the heavy industries	-	50K	Seed	Institutional: Plug and Play Tech Center
93.	iQ3Connect	42	VR-based training and collaboration solution for enterprises	-	1.66M	Seed	Institutional: MassChallenge
94.	Eserv	42	Provider of SaaS-driven asset integrity management solutions for the oil and gas industry	-	-	Seed	Institutional: Vespa Capital
95.	Gleechi	42	Provider of a software for automating hand animation process in games	Minicorn	3.15M	Seed	Institutional: EIC Fund
96.	Imerso	42	Provider of a cloud-based construction monitoring platform	-	2M	Series A	Institutional: Vito One
97.	Immerse	42	Provider of a single VR app for immersive training, offering learning paths that include apps, forms, and gamification	Minicorn	28.1M	Series C	Institutional: Blueray Capital
98.	Pixaera	41	Provider of a VR-based workplace safety training solution for enterprises	Minicorn	5.7M	Series A	Institutional: LocalGlobe

Table 1. Continued

No.	Company name	Tracxn score	Description	X-corn status	Total equity funding (USD)	Company stage	Investor name
98.	Pixaera	41	Provider of a VR-based workplace safety training solution for enterprises	Minicorn	5.7M	Series A	Institutional: LocalGlobe
99.	Resolve	41	Provider of a VR-based collaboration solution for BIM application	-	-	Funding Raised	Institutional: Metaplanet
100.	Voxware	41	Voice recognition software	-	7.85M	Series B	Institutional: Cross Atlantic Capital Partners